

BRS – Advantage Online Shopping

Modules: Login, Shopping Cart, Product Search

1. Purpose

Defines requirements for core modules enabling secure login, product discovery, and cart management.

2. Requirements

Login

- Provide login form with username and password fields.
- Validate credentials and redirect authenticated users to dashboard.
- Display error message for invalid login attempts.
- Provide logout option returning user to homepage.
- Lock account after 5 failed attempts.

Shopping Cart

- Allow adding products from listing or detail pages.
- Display cart icon with item count.
- Show product details (name, price, quantity, subtotal).
- Allow updating/removing items with automatic recalculation.
- Retain cart contents during active session.
- Provide checkout button redirecting to checkout process.

Product Search

- Provide search bar for keywords.
- Display relevant results with product details.
- Allow filtering (category, price, brand) and sorting.
- Handle “No results found” gracefully.
- Retain filters and sorting preferences during session.

3. Non-Functional Requirements

- **Performance:** Login, cart updates, and search results must load within 2 seconds under normal load.

- **Security:** All data transmissions must use HTTPS; prevent injection attacks in search; enforce account logout.
- **Usability:** Forms and navigation must be intuitive and mobile-friendly.
- **Reliability:** System availability must be 99.9% for core modules.
- **Compatibility:** Support major browsers (Chrome, Firefox, Edge, Safari).
- **Scalability:** Must handle concurrent sessions without degradation.